Mozhgan Nasr Azadani

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Mozhgan Nasr | Personal Website | Google Scholar

200 University Ave West, Waterloo, ON, N2L 3G1, Canada

Research Interest

The goal of my research is to develop the algorithmic and representational foundations of multimodal AI systems that integrate vision, language, and action to enable grounded reasoning, predictive world modeling, and reliable decision-making in complex real-world environments.

Education

- **Ph.D. in Computer Science** — *Jan 2019 - Mar 2024*
  University of Ottawa — Ottawa, Canada
  - Advisor: Prof. Azzedine Boukerche
  - Thesis title: "Driving Behavior Analysis and Prediction for Safe Autonomous Vehicles"
  - Nominated for the Best Ph.D. Thesis Award.
- **M.S. in Software Engineering** — *Sep 2015 - Jan 2018*
  Isfahan University of Technology — Isfahan, Iran
  - Advisor: Prof. Nasser Ghadiri
  - Thesis title: "Graph-based biomedical text summarization"
- **B.S. in Computer Science** — *Sep 2009 - Jan 2013*
  University of Isfahan — Isfahan, Iran

Research Experience

- **University of Waterloo** — *May 2024 - Present*
  Postdoctoral Fellow — Waterloo, Canada
  - Advisor: Prof. Krzysztof Czarnecki at WISE Lab.
  - My research focuses on multimodal foundation models, including efficient architectures, grounded reasoning, and reliable decision-making for real-world applications, particularly autonomous driving. Duties also include mentoring graduate students, delivering guest lectures, contributing to research grant proposal writing, and sub-reviewing grant proposals and academic papers.
- **Stanford University** — *Feb 2026 - Aug 2026*
  Visiting Postdoctoral Scholar — Stanford, USA
  - Advisor: Prof. Marco Pavone at ASL Lab.
  - My research focused on grounded multimodal large language models and hallucination mitigation to improve the reliability and interpretability of multimodal reasoning. Duties also include mentoring graduate students.
- **University of Ottawa** — *Jan 2019 - May 2024*
  Graduate Research Assistant — Ottawa, Canada
  - Advisor: Prof. Azzedine Boukerche at PARADISE Lab.
  - Devised a transformer-based algorithm for multimodal motion forecasting under uncertainty.
  - Designed a two-step multimodal goal and trajectory forecasting framework, validated on the Argoverse 1.
  - Created a context-aware, multi-agent motion forecasting model for AV using inverse reinforcement learning (IRL).
  - Created interaction-aware motion forecasting models based on Spatio-Temporal Attention Graph Networks.
  - Designed a multimodal motion planning and trajectory prediction system using Graph Neural Networks.
  - Developed a temporal neural network framework for driver identification through learned driver embeddings.
  - Built a Siamese neural network framework for driver verification and impostor detection.
- **University of Isfahan** — *Jun 2017 - Sep 2017*
  Student Research Intern — Isfahan, Iran
  - Collaborated with a multidisciplinary research team to develop employee health management software and assist in defining the project's research agenda.
  - Designed and implemented a machine learning framework for summarizing Electronic Medical Records (EMRs) using UMLS and the FP-Growth algorithm.
- **Isfahan University of Technology** — *Sep 2015 – Jan 2018*
  Graduate Research Assistant — Isfahan, Iran
  - Designed a graph-based framework for summarizing Electronic Medical Records, achieving improved performance on the ROUGE evaluation metric.

Honors and Awards

- **Globalink Research Award** *Feb 2026 - Aug 2026*
  Mitacs
  ◦ Competitive Research award supporting the international research collaboration.
- **Postdoctoral Fellowship** *2024 - 2026*
  Natural Sciences and Engineering Research Council of Canada (NSERC)
  ◦ Prestigious fellowship supporting outstanding postdoctoral researchers.
- **Merit Scholarship** *2021 – 2023*
  University of Ottawa
  ◦ Awarded in recognition of outstanding academic achievement during doctoral studies.
- **International Doctoral Scholarship** *2019 – 2023*
  University of Ottawa
  ◦ Competitive scholarship awarded to outstanding international Ph.D. students.
- **Excellence in Graduate Studies** *2018*
  Isfahan University of Technology
  ◦ Awarded in recognition of exceptional graduate academic performance.

Teaching

- **Autonomous Vehicles (ECE 495)** *Winter 2025*
  Guest Lecturer, University of Waterloo
  ◦ Delivered two guest lectures on *MLLMs: From Foundations to Driving with Natural Language.*
- **Principles of Intelligent Transportation Systems (CSI 5107)** *Winter 2025*
  Guest Lecturer, University of Ottawa
  ◦ Delivered a guest lecture titled *Driving Behavior Analysis Guidelines for Intelligent Transportation Systems*.
- **Principles of Intelligent Transportation Systems (CSI 5107)** *Winter 2024*
  Guest Lecturer, University of Ottawa
  ◦ Delivered a guest lecture titled *Human Behavior Understanding in Autonomous Driving*.
- **Privacy Enhancing Technologies (CSI 5139)** *Winter 2023*
  Teaching Assistant, University of Ottawa
- **Introduction to Formal Languages (CSI 3104) (×3)** *Spring 2020 - Winter 2023*
  Teaching Assistant, University of Ottawa
- **Cryptography (CSI 4108 / SEC 5101) (×4)** *Fall 2019 - Fall 2022*
  Teaching Assistant, University of Ottawa
- **Natural Language Processing (CSI 5386)** *Winter 2022*
  Teaching Assistant, University of Ottawa
- **Data Mining and Concept Learning (CSI 5387)** *Winter 2020*
  Teaching Assistant, University of Ottawa
- **Advanced Database Systems** *Winter 2016*
  Teaching Assistant, Isfahan University of Technology
- **Database Systems II** *Fall 2016*
  Teaching Assistant, Isfahan University of Technology

Research Mentorship

Provided research/project mentorship to master's and Ph.D. students in multimodal foundation models, machine learning, computer vision, and robotics.

- 1 × Ph.D., University of Toronto *Jul 2026 - Present*
- 1 × M.Sc., 3 × Ph.D., Stanford University *Feb 2026 - Present*
- 5 × M.Sc., 2 × Ph.D., University of Waterloo *Dec 2024 - Present*
- 3 × M.Sc., University of Ottawa *2021 - 2024*

Publications

For a complete and the most up-to-date publication record, please visit my Google Scholar.

Preprints/Under reviews

[P1] A. Jeddi*, E. Zhang*, H. Can Karaimer, J. Gerigk, **M. Nasr Azadani**, J. Luo, M. Ngoc Le, G. Aminian, H. Buurmeijer, I. Gilitschenski, K. G. Derpanis, M. Pavone, B. Taati. SCOPD: Sparse-Context On-Policy Distillation for Efficient Vision Language Models. *under review*, 2026.

[P2] Y. Wang, Y. Choi, B. Zhang, **M. Nasr Azadani**, S. Sedwards, K. Czarnecki. Where Does the Answer Come From? Benchmarking View-Level Visual Evidence Identification in Multi-View MLLMs for Autonomous Driving. *Under review*, 2026.

Journal Articles

[J1] **M. Nasr Azadani**, J. Riddell, S. Sedwards, and K. Czarnecki. Rethinking the Mixture of Vision Encoders Paradigm for Enhanced Visual Understanding in Multimodal LLMs. *Transactions on Machine Learning Research (TMLR), **Awarded J2C certification**, **to be presented at NeurIPS**)*, 2026.

[J2] **M. Nasr Azadani**, and A. Boukerche. Hierarchical Transformers for Motion Forecasting Based on Inverse Reinforcement Learning. *IEEE Transactions on Vehicular Technology (T-VT)*, 2025.

[J3] **M. Nasr Azadani**, and A. Boukerche. CAPHA: A Novel Context-Aware Behavior Prediction System of Heterogeneous Agents for Autonomous Vehicles. *IEEE Transactions on Vehicular Technology (T-VT)*, 2023.

[J4] **M. Nasr Azadani**, and A. Boukerche. STAG: A Novel Interaction-Aware Path Prediction Method Based on Spatio-Temporal Attention Graphs for Connected Automated Vehicles. *Ad Hoc Networks*, 2023.

[J5] **M. Nasr Azadani**, and A. Boukerche. A Novel Multimodal Vehicle Path Prediction Method Based on Temporal Convolutional Networks. *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2022.

[J6] **M. Nasr Azadani**, and A. Boukerche. Siamese Temporal Convolutional Networks for Driver Identification Using Driver Steering Behavior Analysis. *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2022.

[J7] **M. Nasr Azadani**, and A. Boukerche. DriverRep: Driver Identification Through Driving Behavior Embeddings. *Journal of Parallel and Distributed Computing*, 2022.

[J8] **M. Nasr Azadani**, and A. Boukerche. Driving Behavior Analysis Guidelines for Intelligent Transportation Systems. *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2021.

[J9] **M. Nasr Azadani**, N. Ghadiri, and E. Davoodijam. Graph-Based Biomedical Text Summarization: An Itemset Mining and Sentence Clustering Approach. *Journal of Biomedical Informatics*, 2018.

Conference Proceedings

[C1] **M. Nasr Azadani**, Y. Wang, Y. Zhu, L. Chen, M. Ganai, S. Sedwards, M. Pavone, K. Czarnecki. VistaQA: Benchmarking Joint Visual Question Answering and Pixel-Level Evidence. *NeurIPS*, 2026 (Accepted).

[C2] J. Dao*, M. Ganai*, Y. Abukhadra, A. Sridhar, **M. Nasr Azadani**, K. Luo, C. Barrett, J. Wu, C. Finn, M. Pavone. DIRECT: When and Where Should You Allocate Test-Time Compute in Embodied Planners?. *CoRL*, 2026 (Accepted).

[C3] Y. Wang, **M. Nasr Azadani**, S. Sedwards, and K. Czarnecki. HAWAII: Hierarchical Visual Knowledge Transfer for Efficient Vision-Language Models. *NeurIPS*, 2025.

[C4] Y. Wang*, **M. Nasr Azadani***, S. Sedwards, and K. Czarnecki. LEO-MINI: An Efficient Multimodal Large Language Model Using Conditional Token Reduction and Mixture of Multi-Modal Experts. *EMNLP*, 2025.

[C5] **M. Nasr Azadani**, and A. Boukerche. A Novel Transformer-Based Model for Motion Forecasting in Connected Automated Vehicles. *IEEE International Conference on Communications (ICC)*, 2024.

[C6] **M. Nasr Azadani**, and A. Boukerche. GMP: Goal-Based Multimodal Motion Prediction for Automated Vehicles. *IEEE Global Communications Conference (GLOBECOM)*, 2023.

[C7] **M. Nasr Azadani**, and A. Boukerche. A Context-Aware Path Forecasting Method for Connected Autonomous Vehicles. *IEEE International Conference on Communications (ICC)*, 2023.

[C8] **M. Nasr Azadani**, and A. Boukerche. A Social-Aware Vehicle Path Forecasting Method Using Graph Neural Networks. *ACM Symposium on Design and Analysis of Intelligent Vehicular Networks and Applications*, 2023.

[C9] **M. Nasr Azadani**, and A. Boukerche. A Novel Multimodal Behavior Prediction Method for Automated Vehicles. *ACM Symposium on Design and Analysis of Intelligent Vehicular Networks and Applications*, 2023.

[C10] **M. Nasr Azadani**, and A. Boukerche. An Interaction-Aware Vehicle Behavior Prediction for Connected Automated Vehicles. *IEEE International Conference on Communications (ICC)*, 2022.

[C11] **M. Nasr Azadani**, and A. Boukerche. An Internet-of-Vehicles Powered Defensive Driving Warning Approach for Traffic Safety. *IEEE Global Communications Conference (GLOBECOM)*, 2021.

[C12] **M. Nasr Azadani**, and A. Boukerche. Toward Driver Intention Prediction for Intelligent Vehicles: A Deep Learning Approach. *IEEE Conference on Local Computer Networks (LCN)*, 2021.

[C13] **M. Nasr Azadani**, and A. Boukerche. Driver Identification Using Vehicular Sensing Data: A Deep Learning Approach. *IEEE Wireless Communications and Networking Conference (WCNC)*, 2021.

[C14] **M. Nasr Azadani**, and A. Boukerche. Performance Evaluation of Driving Behavior Identification Models Through CAN-BUS Data. *IEEE Wireless Communications and Networking Conference (WCNC)*, 2020.

[C15] **M. Nasr Azadani**, and N. Ghadiri. Evaluating Different Similarity Measures for Automatic Biomedical Text Summarization. *Intelligent Systems Design and Applications*, 2017.

[C16] E. Maleki*, **M. Nasr Azadani***, and N. Ghadiri. Performance Evaluation of SpatialHadoop for Big Web Mapping Data. *International Conference on Web Research*, 2016.

Workshop Papers

[W1] J. Dao*, M. Ganai*, Y. Abukhadra, A. Sridhar, **M. Nasr Azadani**, K. Luo, C. Barrett, J. Wu, C. Finn, M. Pavone. DIRECT: When and Where Should You Allocate Test-Time Compute in Embodied Planners? *RSS FM4RoboPlan* (**Oral Presentation**) and *ECCV EMR*, 2026.

[W2] A. Jeddi*, E. Zhang*, H. Can Karaimer, J. Gerigk, **M. Nasr Azadani**, J. Luo, M. Ngoc Le, G. Aminian, H. Buurmeijer, I. Gilitschenski, K. G. Derpanis, M. Pavone, B. Taati. SCOPE: Sparse-Context On-Policy Distillation for Efficient Vision Language Models. *NeurIPS VLM4RWD*, 2026.

[W3] Y. Wang, **M. Nasr Azadani**, X. Liu, Y. M. Choi, B. Xue, K. Czarnecki. Findings of the First GroundLM Shared Tasks: Evaluating Grounded Language Models across Visual and Scientific Evidence. *EMNLP GroundLM Workshop*, 2026.

[W4] **M. Nasr Azadani**, and A. Boukerche. Convolutional and Recurrent Neural Networks for Driver Identification: An Empirical Study. *IEEE NOMS Workshop on Intelligent Transportation and Autonomous Vehicle Technologies*, 2022.

Invited Talks

- **Mixture of Vision Encoders for MLLMs** *Mar 2026*
  Stanford University, CA, USA
- **Toward Efficient and Reliable VLMs for Real-World Autonomous Systems** *Nov 2025*
  Workshop on Embodied and Safe-Assured Robotic Systems, NeurIPS 2025
- **Driving Behavior Analysis Guidelines for Intelligent Transportation Systems** *Feb 2023*
  Principles of Intelligent Transportation Systems, University of Ottawa, ON, Canada
- **Graph-based Biomedical Text Summarization** *Nov 2017*
  Isfahan University of Technology, Isfahan, Iran

Workshop Organization

- **2nd workshop on Grounded and Faithful Vision-Language Models for Real-World Deployment** *Dec 2026*
  NeurIPS 2026 VLM4RWD (lead organizer)
  ◦ Focused on grounded understanding and faithful reasoning of VLMs/VLAs
- **1st workshop on Grounding Language Models: Learning Faithfully and Efficiently** *Oct 2026*
  EMNLP 2026 GroundLM
  ◦ Focused on grounding, faithfulness, and efficiency in large language models
- **1st workshop on how to Build Effective World Models for Embodied AI** *Sep 2026*
  ECCV 2026 WMEAI
  ◦ Focused on world models integration into embodied AI systems
- **1st workshop on Vision Language Models: Challenges of Real World Deployment** *Nov 2025*
  NeurIPS 2025 VLM4RWD (lead organizer)
  ◦ Focused on efficiency, robustness, and reliability of vision language models

Academic Service

- **Area Chair** *2026*
  NeurIPS 2026
- **Journal Reviewer** *2022 - present*
  I am an active reviewer for the following venues:
  ◦ Transactions on Machine Learning Research (TMLR)
  ◦ IEEE Transactions on Vehicular Technology (T-VT)
  ◦ IEEE Transactions on Intelligent Vehicles (T-IV)
  ◦ IEEE Transactions on Intelligent Transportation Systems (T-ITS)
  ◦ IEEE Transactions on Neural Networks and Learning Systems (T-NNLS)
  ◦ Nature Communications
  ◦ IEEE Transactions on Multimedia
  ◦ Pattern Recognition
  ◦ IEEE Access
  ◦ IEEE Transactions on Mobile Computing

- IEEE Transactions on Consumer Electronics
- Applied Artificial Intelligence
- The Open Transportation Journal

- **Conference Reviewer** *2023 - present*
  I have served as a reviewer for the following venues:
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
  - IEEE/CVF International Conference on Computer Vision (ICCV)
  - Annual Meeting of the Association for Computational Linguistics (ACL)
  - Conference on Neural Information Processing Systems (NeurIPS)
  - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
  - IEEE International Conference on Communications (ICC)
  - IEEE Wireless Communications and Networking Conference (WCNC)
  - IEEE Global Communications Conference (GLOBECOM)
  - IEEE Intelligent Vehicles Symposium (IV)
  - IEEE International Conference on Distributed Smart Cameras (DSC)

OTHER ACTIVITIES

- **IEEE Intelligent Transportation Systems Society member** *2023 – Present*
  ITSS
- **IEEE Women in Engineering (WIE) Member** *2021 – Present*
  IEEE
  - Participated in professional development, networking, and leadership events, including the IEEE WIE Leaders Forum organized by IEEE Region 7 and the Ottawa Section WIE.
- **Institute of Electrical and Electronics Engineers Member** *2021 – Present*
  IEEE
- **IEEE Young Professionals member** *2021 – Present*
  IEEE
- **Iranian Student Association of University of Ottawa** *2020 – 2022*
  Volunteer
  - Organized and delivered webinars and orientation sessions for incoming students.